Multimodal Insurance Creator

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number: 09D6

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Academic Year: 2025-2026

Problem Statement & Objectives

1. Problem Statement

Understanding insurance concepts such as risk, coverage, and policy benefits can be complex and difficult to visualize. Traditional methods rely heavily on textual explanations, which may not effectively communicate these concepts to users.

The Multimodal Insurance Creator aims to solve this problem by leveraging Generative AI to transform textual prompts into both descriptive narratives and visual representations, enabling better comprehension and engagement.

2. Project Objectives

- To design and develop a multimodal AI-based system that generates both textual and visual outputs from a single input
- To integrate Large Language Models (LLMs) for generating structured and meaningful explanations
- To incorporate image generation models for producing visual representations such as infographics
- To implement parallel processing to improve system efficiency and response time
- To build a modular and scalable architecture that supports future enhancements
- To provide a user-friendly web interface for seamless interaction
- To demonstrate real-world application of Generative AI in the insurance domain

3. Scope of the Project

The scope of the project defines the boundaries and capabilities of the system. It focuses on creating a real-time multimodal AI application for insurance concept visualization.

- Covers concepts like risk, coverage, and policies
- Generates multimodal outputs (text + image)
- Uses external AI APIs for processing
- Supports real-time interaction via web interface
- Implements stateless backend design
- Allows future expansion to other domains

Proposed Solution

Key features

The system is designed with several core features that enable efficient multimodal content generation and user interaction.

- Multimodal output generation (text + image)
- Parallel execution of AI services
- API-based communication architecture
- Structured and user-friendly output display
- Scalable and modular design
- Real-time processing capability
- Easy deployment using containerization

Overall Architecture / Workflow

The system follows a layered and modular architecture to ensure separation of concerns, scalability, and maintainability.

1. Presentation Layer (Frontend)

- Provides a web-based interface
- Captures user input prompts
- Displays generated text and images in a structured format

2. Application Layer (Backend)

- Acts as the central controller of the system
- Handles request validation and processing
- Manages communication between frontend and AI services
- Executes parallel processing of AI requests

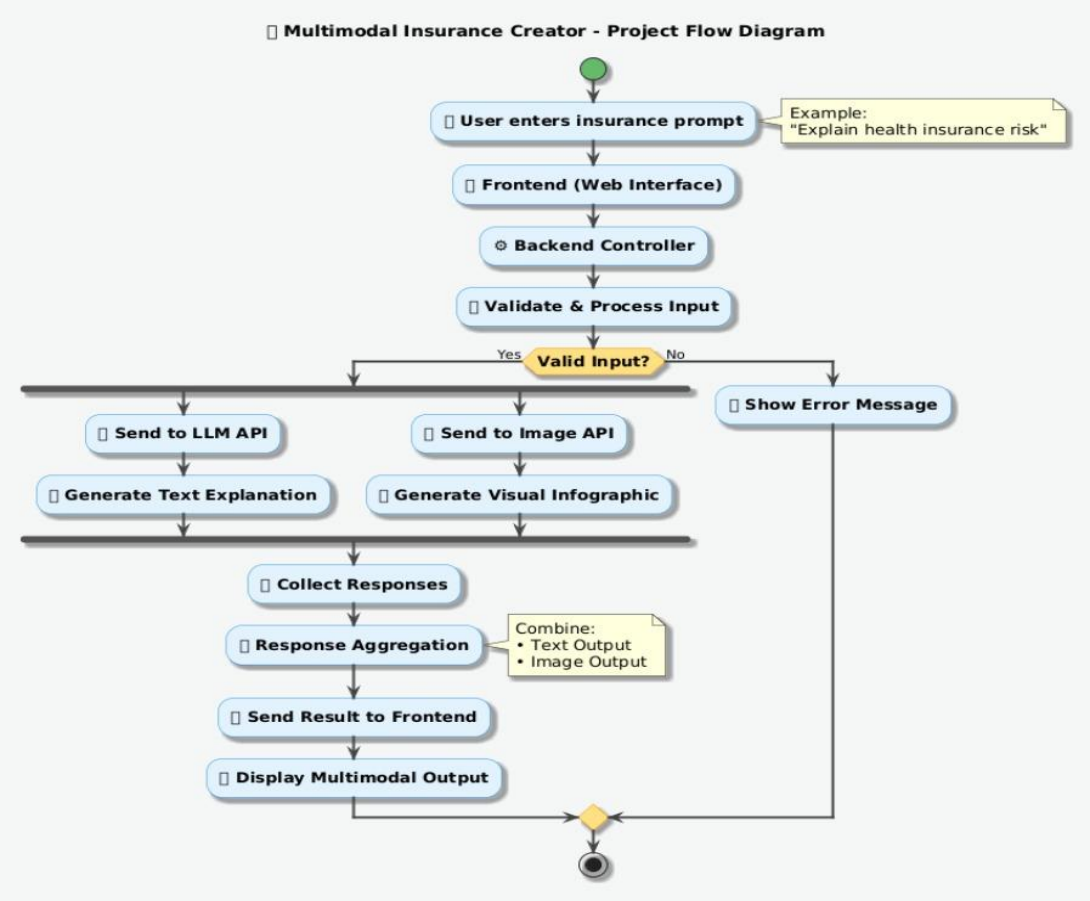
3. AI Service Layer

- **LLM API:** Generates descriptive textual output
- **Image Generation API:** Produces visual representations such as infographics

Workflow Explanation:

1. The user enters an insurance-related prompt via the interface
2. The frontend sends the request to the backend
3. The backend validates and preprocesses the input
4. The processed input is sent simultaneously to:
    - LLM service (text generation)
    - Image generation service (visual output)
5. Both services process the request in parallel
6. The backend collects responses from both services
7. Outputs are aggregated into a unified response
8. The final result is displayed to the user

This approach improves performance, responsiveness, and user experience while maintaining system flexibility.



Tools & Technologies Used

Programming & Frameworks

- Python
- Flask (Backend Framework)

AI & Data Technologies

- LLM API (for text generation)
- Image Generation API (for visuals)
- Vector Database (for embeddings and contextual understanding)

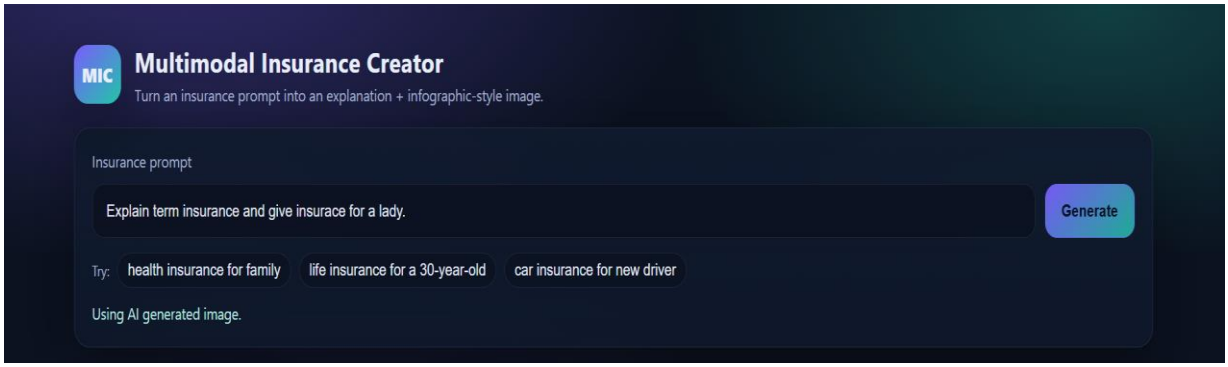
Data Handling

- JSON for API communication

GitHub (Version Control & Collaboration)

Results & Output

1.Screenshots



Explanation

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**Term Insurance: A Beginner's Guide**

### What is Term Insurance?

Term insurance is a type of life insurance that provides coverage for a
specific period (the term) for a premium paid. If the policyholder dies
within the term, the insurance company pays a death benefit to the
beneficiary. If the policyholder survives the term, the coverage ends,
and no payout is made.

**Key Features:**

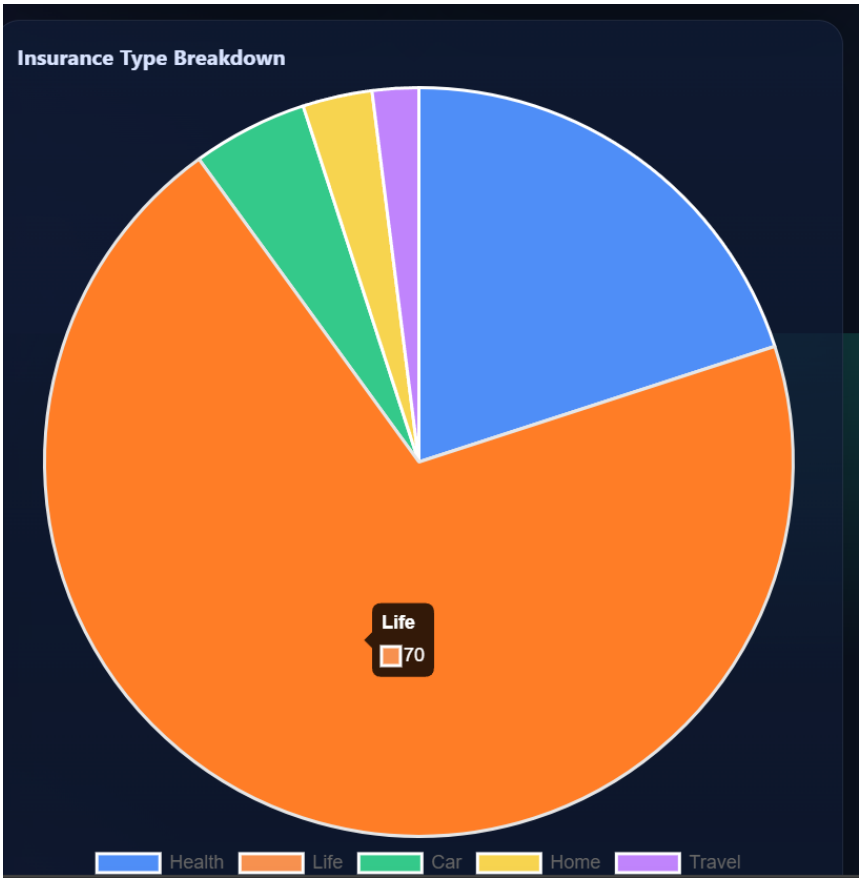
* Provides coverage for a specific period (e.g., 10, 20, or 30 years)
* No cash value accumulation
* Premiums are typically lower than permanent life insurance
* Coverage ends at the end of the term

### Benefits of Term Insurance

* **Affordable premiums**: Term insurance is generally less expensive
than permanent life insurance
* **Flexibility**: Can be converted to a permanent policy or increased in
coverage
* **Tax-free benefits**: Death benefits are tax-free to the beneficiary
* **Simplified underwriting**: Easier to qualify for coverage
```

Visual

Visual Breakdown
Rating
4.8 / 5
Monthly premium (est.)
₹ 515
Premium vs Coverage
6,00,000
5,00,000
4,00,000
3,00,000
2,00,000
1,00,000
0
Term 10
Term 20
Term 30
Annual Premium (₹)
Coverage (₹)


Insurance Type Breakdown
Life
70
Health
Life
Car
Home
Travel

Key Outcomes

• The successful implementation of this project highlights the effectiveness of combining multiple AI technologies into a single system.

- Developed a working multimodal AI application
- Achieved parallel execution of AI services
- Built scalable and modular architecture
- Improved understanding of insurance concepts
- Demonstrated real-world GenAI usage
- Efficient integration of APIs and backend

Conclusion

The Multimodal Insurance Creator project effectively demonstrates the integration of multiple Generative AI technologies to create a unified system capable of producing both textual and visual outputs. By combining LLM-based text generation with image synthesis models, the system enhances the way complex insurance concepts are understood and presented.

The project also highlights key software engineering principles such as modularity, scalability, API-driven architecture, and parallel processing, making it a strong example of a real-world GenAI application. Overall, the system provides an innovative and efficient solution for interactive concept visualization.

Future Scope & Enhancements

The project can be further improved by adding advanced features and expanding its capabilities to support more complex use cases.

- Integration of audio and video generation models
- Implementation of fine-tuned domain-specific LLMs
- Addition of user authentication and history tracking
- Deployment using Kubernetes for large-scale scalability
- Enhancement of UI/UX for better user experience
- Multi-language support for wider accessibility
- Integration with real-time analytics dashboards
- Improved prompt engineering for higher-quality outputs